

Use computer vision for real-time surface defect detection in manufacturing, addressing cold-start with few-shot learning.

Manufacturing × Automated quality inspection × Computer vision · OS008 v4 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
48 is the world moving	52 can we play, can we win	MEDIUM 0 named in the evidence	€107m observed evidence · per year	LATER research and standards activi...	L2 23 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

We help manufacturers deploy real-time computer vision for surface defect detection that works from day one, even with limited defect samples. By combining few-shot learning with edge-optimized models, we overcome the cold-start problem that stalls traditional inspection systems. This matters now because recent advances in vision-language models and edge AI make it practical to deploy on the factory floor.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€139m €25.0m – €746m	€107m €19.2m – €572m	€799k €144k – €4.3m	observed
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€0	€0	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (DE)	51,224 enterprises	observed	Eurostat · sbs_sc_oww	2024
Enterprises adopting Computer vision	4.0 % of enterprises (10+ employees)	observed	Eurostat · isoc_eb_ain2 · E_AI_TIR	2025
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	10,301 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	0.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 3 of 4 declared geographies are outside the European reference data (CN, IN, US); the estimate covers the European footprint only.

Observed: tendered contract value, annualised

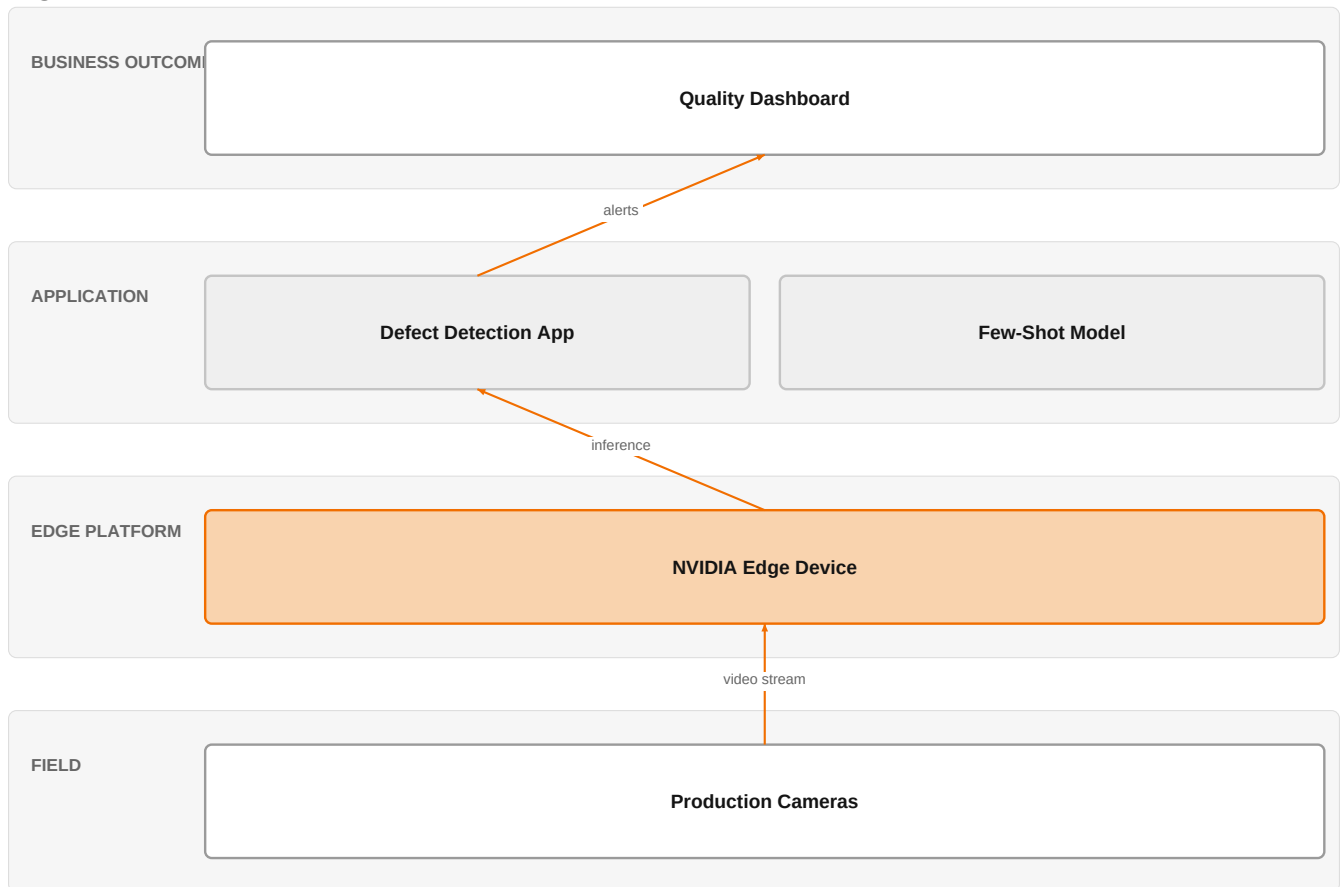
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	0.8 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size. 3 of 4 declared geographies are outside the European reference data (CN, IN, US); the estimate covers the European footprint only.

The solution, in one picture

Edge AI Defect Detection



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This picture shows how a few-shot defect detection model runs on edge hardware to provide real-time quality alerts without sending data to the cloud.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Recent research shows that vision-language models like YOLO-World can be adapted for few-shot industrial defect detection, directly addressing the cold-start problem of limited annotated data. Edge-optimized detection frameworks, such as a lightweight YOLOv11 variant, enable real-time safety and quality decisions on resource-constrained industrial devices. Additionally, systematic reviews highlight that computer vision and machine learning are increasingly applied to defect detection in materials like 3D-printed cementitious elements and textiles, where manual inspection is inefficient. These advances make it feasible to deploy automated inspection that is both accurate and low-latency in real production environments.

Sources: SIG-8CA4FFD436BC openalex.org 2026-03-27; SIG-5B350B1A3B15 Journal of Organizatio 2026-05-21; SIG-AE9A7F9C36FF openalex.org 2026-05-04; SIG-CF16251C55F0 openalex.org 2025-11-02

Who buys, and why now

The head of manufacturing or quality assurance signs the contract, but the production line manager feels the pain of missed defects and slow manual inspection. The trigger is often a costly recall or a customer complaint about quality, or a new production line that needs inspection from day one without months of data collection. They need to reduce scrap, avoid downtime, and meet quality standards without hiring more inspectors.

Why it is hot now — every claim bound to a dated source

- Computer vision and machine learning techniques are being used to identify surface defects in 3D-printed cementitious materials. [SIG-AE9A7F9C36FF]

- Vision-language models adapted for few-shot industrial defect detection address data scarcity. [SIG-8CA4FFD436BC]
- Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing uses deep learning. [SIG-A598AE09F54F]
- Deep learning methods enhance textile and colour defect detection. [SIG-CF16251C55F0]
- Machine vision-based intelligent algorithms are applied to gear defect detection. [SIG-E9AF6D858220]
- Vision-based quality monitoring with deep learning detects defects in large-scale earthen 3D printing in real time. [SIG-A598AE09F54F]
- Automated surface defect detection in 3D-printed cementitious materials is achieved using computer vision and machine learning. [SIG-AE9A7F9C36FF]
- Machine vision-based gear defect detection supports quality control in intelligent manufacturing, leveraging non-contact measurement and high efficiency. [SIG-E9AF6D858220]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Orange Group researchers to design and deploy a computer vision pipeline. The engagement starts with a pilot on one production line, using NVIDIA edge hardware (partner) to run a few-shot learning model that can be trained on a handful of defect images. Orange's ISO 27001 and TISAX certifications provide the trust framework for handling production data. The team integrates the model with existing cameras and PLCs, and provides a dashboard for real-time alerts. Finally, they scale to multiple lines, using federated learning to improve the model across sites without centralizing data.

Why Orange can win

Orange brings a unique combination of industrial IoT expertise, research capability, and certified security (ISO 27001, TISAX) that pure software vendors lack. The partnership with NVIDIA gives access to optimized edge hardware, while the reference implementations at Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass demonstrate real-world deployment. However, the specific computer vision assets are thin—Orange does not have a proprietary detection model, so the win depends on integration and trust rather than a unique algorithm.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Computer vision	technology	L3	no offer path, but adjacent assets or Orange research exist	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 5.2; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
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Bosch	Industrial / OT specialist	sells Computer vision; active in Manufacturing; competes in OX: Smart Industries	structural	—
Dedrone by Axon	Security specialist	sells Computer vision; competes in OX: Smart Industries	structural	—
HENSOLDT	Security specialist	sells Computer vision	structural	—
Honeywell	Industrial / OT specialist	sells Computer vision; active in Manufacturing; competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 How many defect types do you currently inspect for, and how many annotated images do you have per defect type?
- 2 What is your current false positive rate in manual inspection, and what is the cost of a missed defect reaching a customer?
- 3 Do you have existing cameras on the line, or would we need to install new ones?
- 4 What is your tolerance for latency between image capture and defect alert?
- 5 Are you open to running models on edge devices, or do you require cloud-based processing?
- 6 Which production lines are you prioritizing for automated inspection, and what is the expected ramp-up time?

Objections you will meet

They will say	You can answer
We already have a quality inspection system from Bosch/Honeywell.	That's true, and those systems are strong. But if you're seeing cold-start issues or want to add new defect types quickly, our few-shot approach can complement or replace parts of that system with less data and faster deployment.
We don't have enough defect images to train a model.	That's exactly the problem we solve. Using few-shot learning, we can get a working model with just a handful of examples per defect, and then improve it over time as you collect more data.
We're concerned about data privacy and security on the factory floor.	We understand. That's why we bring ISO 27001 and TISAX certifications, and we can deploy models at the edge so your images never leave your facility unless you want them to.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a manufacturing client's automated quality inspection tender, anchoring on our computer vision capability, IoT experts, and ISO 27001, and referencing Saint-Gobain Glass as a relevant reference.
Sales	In a conversation with a manufacturing quality director, say: 'We see manufacturers struggling with defect detection when they lack enough defect images to train a model. We're exploring few-shot learning so you can get value from a small set of examples, and we can tap into our IoT and research teams to tailor it to your line.'
Strategist / Innovator	Commission a deep-dive brief on few-shot learning for industrial surface defect detection, focusing on vision-language models and edge deployment, to decide whether to invest in a manufacturing proof-of-concept next quarter.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-66AAACE64906A	2026-06-04	nist.gov	1	NIST Researchers Discover a New Way to Whisk Alloys Together With Lasers
SIG-5B350B1A3B15	2026-05-21	Journal of Organizational...	1	Latency Trade-Offs of a Lightweight YOLOv11 Detection Algorithm on Edge Computing Nodes for Fac...
SIG-AE9A7F9C36FF	2026-05-04	openalex.org	1	Computer Vision and Machine Learning Approaches for Defect Detection in 3D-Printed Cementitious...
SIG-8CA4FFD436BC	2026-03-27	openalex.org	1	Adapting Vision–Language Models for Few-Shot Industrial Defect Detection
SIG-EB5664191E1A	2026-02-01	openalex.org	1	A Systematic Review of Anomaly and Fault Detection Using Machine Learning for Industrial Machin...
SIG-1D3047429B46	2025-12-12	openalex.org	1	A comprehensive survey for real-world industrial surface defect detection: Challenges, approach...
SIG-92E2E10DA219	2025-12-12	openalex.org	1	A real-time surface defect detection model based on adaptive feature information selection and ...
SIG-E0AA4F59988C	2025-12-10	openalex.org	1	Anomaly detection for industrial applications: challenges, solutions, and future directions
SIG-A598AE09F54F	2025-11-20	openalex.org	1	Real-time vision-based defect detection for large-scale on-site earthen additive manufacturing:...
SIG-247B09FE185A	2025-11-07	openalex.org	1	Advancements in Small-Object Detection (2023–2025): Approaches, Datasets, Benchmarks, Applicati...
SIG-CF16251C55F0	2025-11-02	openalex.org	1	Textile and colour defect detection using deep learning methods
SIG-40D3557E680A	2025-10-23	openalex.org	1	Edge-Optimized Hybrid DNNs for Efficient High-Dimensional Visual Anomaly Detection in Industria...
SIG-E9AF6D858220	2025-10-21	openalex.org	1	Review on Application of Machine Vision-Based Intelligent Algorithms in Gear Defect Detection
SIG-1EDDEA256F85	2025-10-16	openalex.org	1	Evaluation of computer vision techniques for quality inspection in casting manufacturing process
SIG-3C2F7CAC7510	2025-10-01	openalex.org	1	Federated multi scale vision transformer with adaptive client aggregation for industrial defect...
SIG-17D5A7F00D3D	2025-09-25	openalex.org	1	Vision transformer and Mamba-attention fusion for high-precision PCB defect detection
SIG-66E8F1ED640C	2025-09-18	openalex.org	1	Textile Defect Detection Using Artificial Intelligence and Computer Vision—A Preliminary Deep L...
SIG-D46EBCBB2C04	2025-09-16	openalex.org	1	Embedded vision transformers for industrial welding defect detection
SIG-174AD17494CC	2025-09-12	openalex.org	1	Transformer-based Sequential Contrastive Learning for Industrial Robot Time Series Anomaly Dete...
SIG-E4CC78DB859B	2025-09-06	openalex.org	1	System-level integration of deep learning and computer vision for contact ring seal defect dete...
SIG-5DD84E684B5F	2025-08-21	openalex.org	1	Autonomous printing process optimisation and in-situ anomaly detection in fused deposition mode...
SIG-1761232FFDC4	2026-08-06	arxiv.org	3	A Low-Latency ASIC Architecture for Real-Time Line Segment Detection
SIG-A06EDB8B4416	2026-04-22	arxiv.org	3	IoT-Enhanced CNN-Based Labelled Crack Detection for Additive Manufacturing Image Annotation in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); risks_and_unknowns (missing from the model output); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:13+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:16+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:40:41+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.